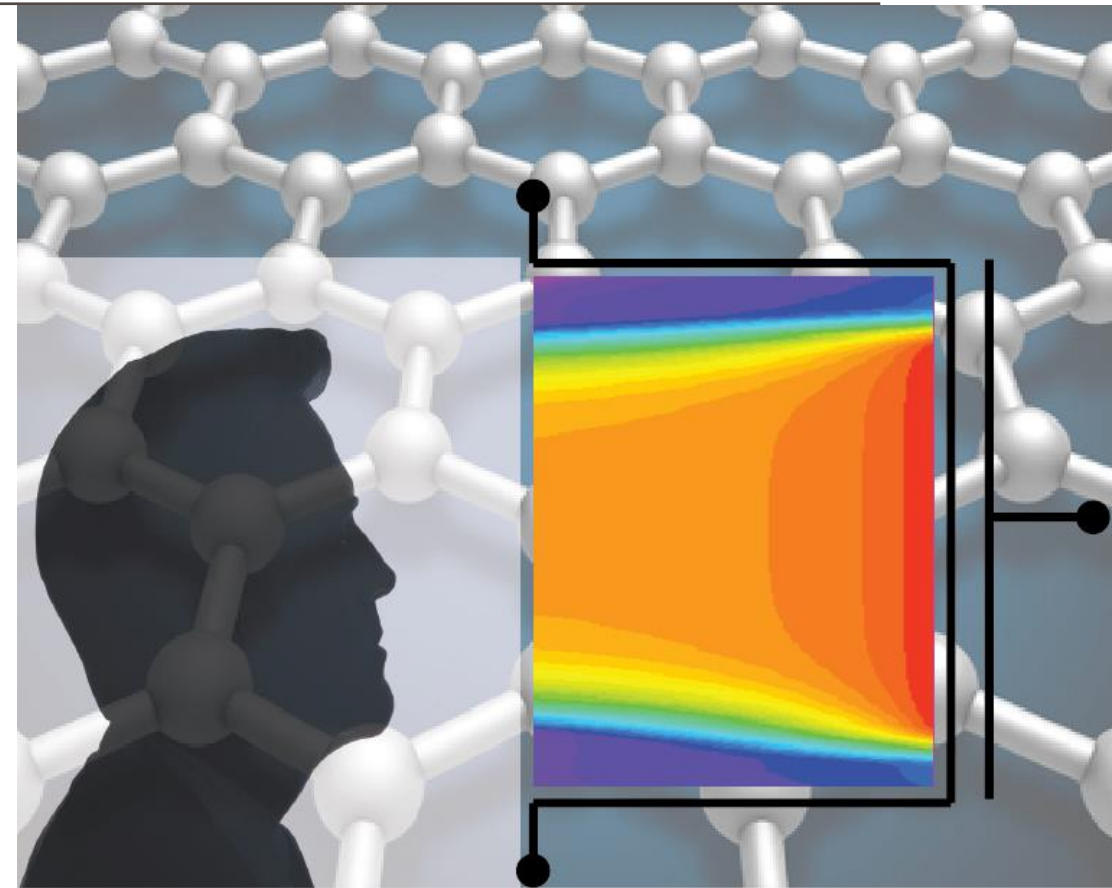


ADVANCED SOLID-STATE DEVICES

Week-2, Lecture-2

Sneh Saurabh
17th January, 2025



Density of States (DOS)

DOS: Definition(2)

- DOS represents if we increase the energy by amount dE , how many states would be covered:

$$D(E)dE$$

⇒ If we integrate $D(E)$ up to an energy E , we should get the number of states less than an energy E

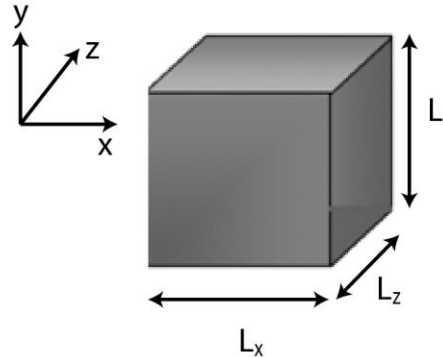
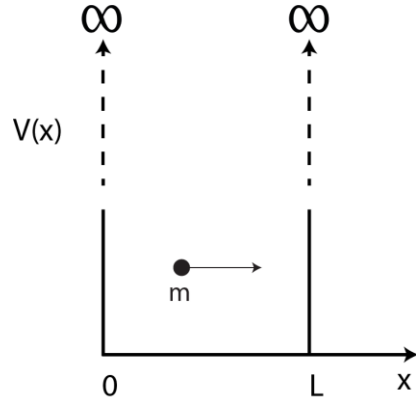
$$N(E) = \int_{-\infty}^E D(E') dE'$$

[$N(E)$ represents the total number of states with energy less than or equal to E]

$$\Rightarrow D(E) = \frac{dN(E)}{dE}$$

- We can use the above relation to compute density of states

DOS Derivation: Spacing between states



- Consider an electron confined in a box of dimension is $L_x \times L_y \times L_z$

$$\text{Energy is quantized } E_n = \frac{n^2 \pi^2 \hbar^2}{2mL^2}$$

- Consider the momentum space
 - p_x , p_y , and p_z along three axes
- The spacing between two momentum states along:
 - X direction will be $\frac{h}{L_x}$
 - Y direction will be $\frac{h}{L_y}$
 - Z direction will be $\frac{h}{L_z}$

DOS Derivation: Counting States

- Aim is to find DOS
- First we find the number of states less than a given momentum p

How to find the number of states less than a given momentum p ?

- Consider a sphere of radius p (in the momentum space)
 - Total volume $\frac{4}{3}\pi p^3$
- Volume of each state = $\frac{h}{L_x} \times \frac{h}{L_y} \times \frac{h}{L_z}$
- Number of states less than the momentum p is $N(p) = \frac{\frac{4}{3}\pi p^3}{(h/L_x)(h/L_y)(h/L_z)} = \frac{4}{3}\pi L_x L_y L_z \left(\frac{p}{h}\right)^3$

DOS Derivation: Recipe

Density of States (DOS):

$$D(E) = \frac{dN(E)}{dE}$$

where $N(E)$ represents the total number of states with energy less than or equal to E

1. We have already computed $N(p)$ (number of states less than or equal to a given momentum p)
2. Using $N(p)$, we can obtain $N(E)$ applying dispersion relation $[E - k$ relation, and using $p = \hbar k]$
3. We can differentiate $N(E)$ with respect to E to obtain $D(E)$ or DOS

DOS Derivation: Conduction band (1)

Problem: Find the DOS $D(E)$ for a 3-D bulk material for which the following parabolic dispersion relation can be assumed in the conduction band:

$E = E_c + \frac{\hbar^2 k^2}{2m^*}$ where m^* is the effective mass of electrons in the conduction band

We have already computed:

$$N(p) = \frac{4}{3} \pi L_x L_y L_z \left(\frac{p}{\hbar} \right)^3$$

$$p = \hbar k = \sqrt{2m^*(E - E_c)}$$

$$N(E) = \frac{4\pi L_x L_y L_z [2m^*(E - E_c)]^{3/2}}{3h^3}$$

Answer:

$$D(E) = \frac{4\pi L_x L_y L_z m^* \sqrt{2m^*(E - E_c)}}{h^3}$$

If we consider 2 electrons occupies each state (with spin +1/2 and -1/2,) then DOS for electrons in conduction band is:

$$D(E) = \frac{8\pi L_x L_y L_z m^* \sqrt{2m^*(E - E_c)}}{h^3}$$

DOS Derivation: Conduction band (2)

- Density of states (denoted as $g(E)$) is often taken in books as $D(E)$ per unit volume
- With this definition of density of state, $g(E)$ becomes a double density, i.e., number per unit energy per unit volume
- Density of state in conduction band $g_c(E) = \frac{D(E)}{L_x L_y L_z} = \frac{8\pi m^* \sqrt{2m^*(E-E_c)}}{h^3}$

DOS Derivation: Valence band (1)

Problem: Find the DOS $D(E)$ for a 3-D bulk material in valence band for which the following parabolic dispersion relation can be assumed:

$$E = E_V - \frac{\hbar^2 k^2}{2m^*} \text{ where } m^* \text{ is the effective mass of holes in the valence band}$$

Answer: (and taking spin into account)

$$D(E) = \frac{8\pi L_x L_y L_z m^* \sqrt{2m^*(E_V - E)}}{h^3}$$

- Density of state in valence band $g_v(E) = \frac{D(E)}{L_x L_y L_z} = \frac{8\pi m^* \sqrt{2m^*(E_V - E)}}{h^3}$

DOS: Conduction and Valence band

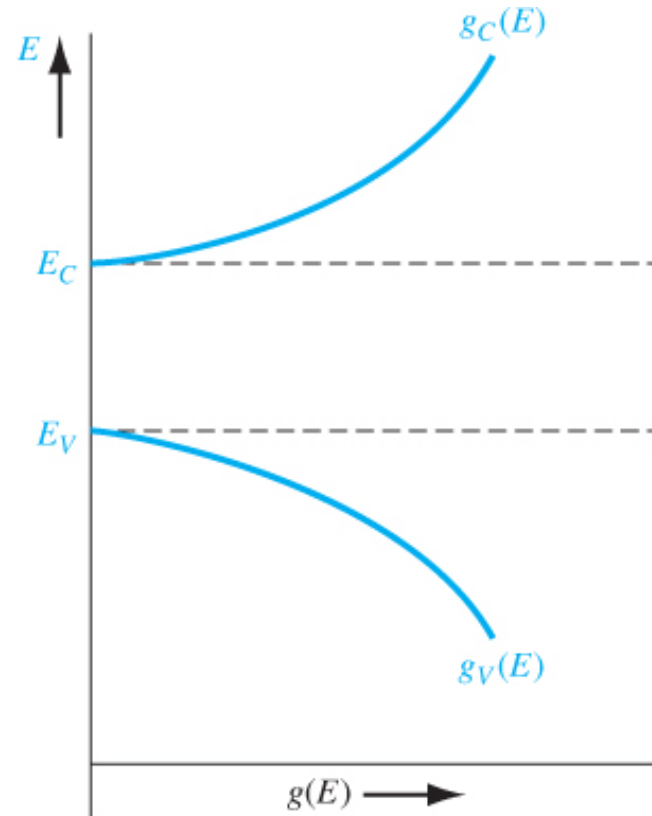


Figure 3.27 | The density of energy states in the conduction band and the density of energy states in the valence band as a function of energy.

Fermi-Dirac Distribution Function

Fermi Distribution Function: Basics

To find how the electrons are distributed among various energy states at **thermal equilibrium**:

1. How many states are available in the given energy interval? **DOS**
2. How are these states occupied by the electrons? **Fermi Distribution Function**

Fermi Distribution Function: probability that an energy level E is occupied by an electron at thermal equilibrium:

$$f(E)$$

Density of States: number of states in an interval E and $E + dE$:

$$D(E)dE$$

Number of electrons: number of electrons in an interval E and $E + dE$:

$$dN = f(E)D(E)dE$$

Fermi Distribution Function: Assumptions

How does the Fermi Distribution Function $f(E)$ look like?

Actual Derivations need understanding of *Statistical Thermodynamics* (we will not do in this course)

Assumptions used in deriving Fermi Distribution Function:

1. Total *number of electrons* (N) and *total energy of the system* (E) have a fixed value for a Fermi Distribution Function
2. **Pauli Exclusion Principle:** Only one electron can occupy a given quantum state
3. Number of electrons is very large

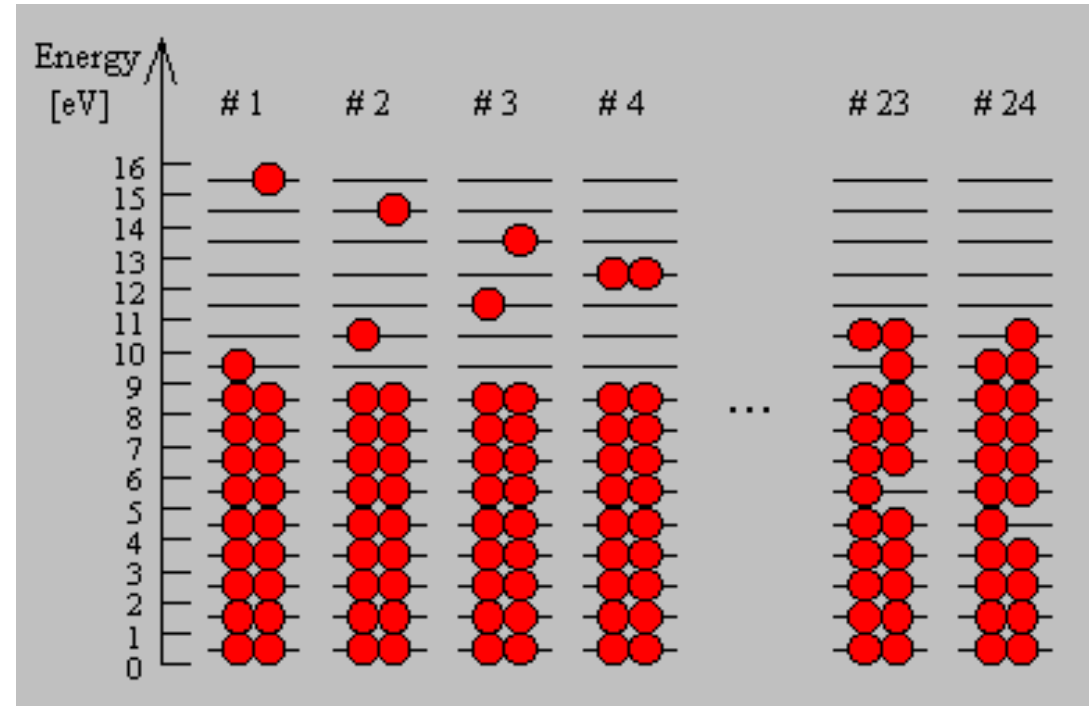
Fermi Distribution Function: Example (1)

Consider a systems with:

- Equidistant energy levels (16 levels):
 $0.5, 1.5, 2.5, 3.5, \dots, 14.5, 15.5 \text{ eV}$
- $N = 20$ electrons
- Total energy of the system $E = 106 \text{ eV}$

What are different ways in which the energy levels can be occupied by $N = 20$ electrons such that total energy of the system is $E = 106 \text{ eV}$?

There are 24 possible configurations



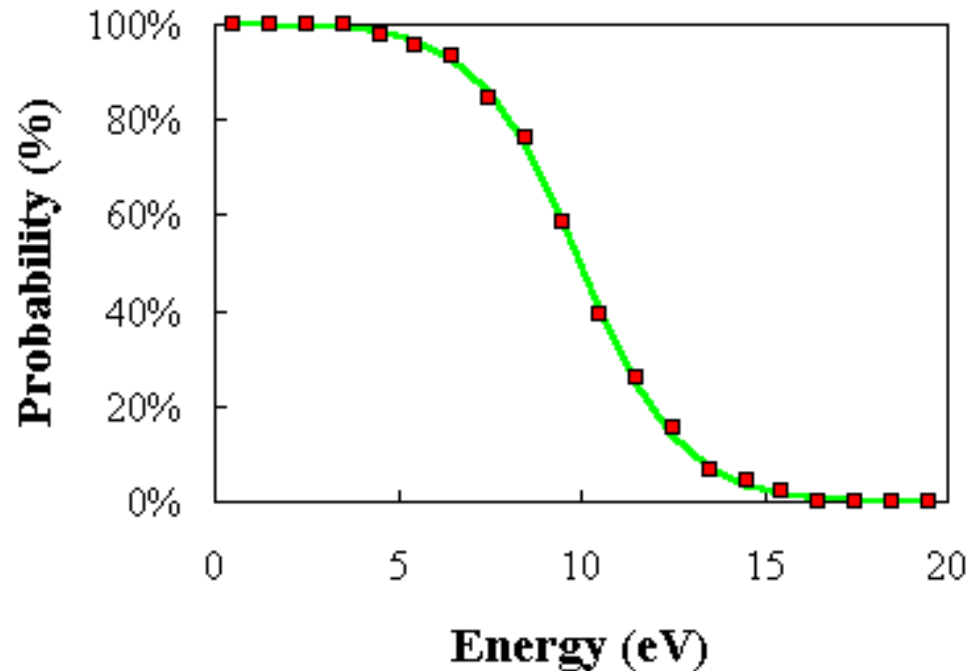
Source: <http://ecee.colorado.edu/~bart/book/distrib.htm>

Fermi Distribution Function: Example (2)

Energy	#1	#2	#3	#4	#5	#6	#7	#8	#9	#10	#11	#12	#13	#14	#15	#16	#17	#18	#19	#20	#21	#22	#23	#24
15.5	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
14.5	0	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
13.5	0	0	0	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
12.5	0	0	0	0	0	0	2	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0
11.5	0	0	0	1	0	0	0	1	0	0	0	0	2	2	2	1	1	1	1	1	0	0	0	0
10.5	0	1	0	0	1	0	0	0	2	1	1	0	1	0	0	2	2	1	1	0	2	2	2	1
9.5	1	0	2	0	1	2	0	1	0	2	1	2	0	2	1	1	0	2	1	2	2	2	1	2
8.5	2	2	1	2	1	2	2	1	1	0	2	2	1	0	2	0	2	1	2	2	2	1	2	2
7.5	2	2	2	2	2	1	2	2	2	2	1	2	2	2	1	2	1	1	2	2	0	2	2	2
6.5	2	2	2	2	2	2	2	2	2	2	2	1	2	2	2	2	2	2	1	2	2	1	2	2
5.5	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	1	2	2	1	2
4.5	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	1
3.5	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2
2.5	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2
1.5	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2
0.5	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2

24 possible configurations

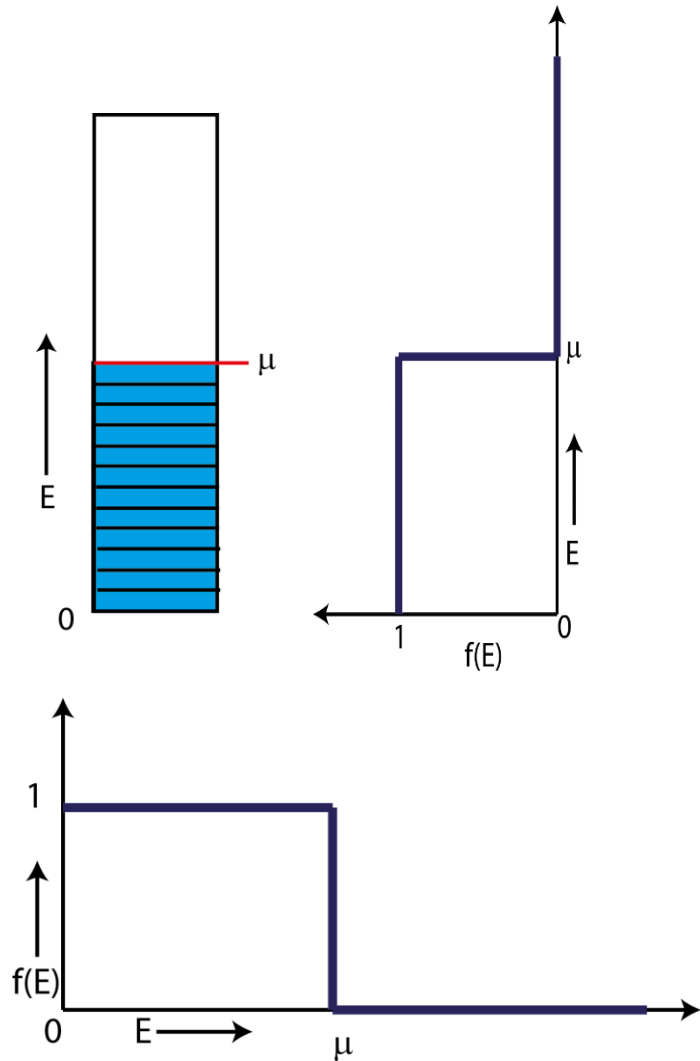
Fermi Distribution Function: Example (3)



Probability of occupation at each energy level

- Fit a curve at all the points
- When N is large this fit would be Fermi Distribution Function
- In this case green curve fits a Fermi Distribution Function (parameters of Fermi Distribution Function has been chosen to give a good fit)
- Considering that the system is composed of small number of electrons, very good fit is surprising (and may not be true in general)

Fermi Distribution Function: At 0K



- Electrons are filled starting with the lowest Energy Level
- Pauli Exclusion Principle stops electrons from all getting into the lowest energy state

- All states below a certain energy level μ , called *Fermi Energy Level* or *Electrochemical Potential*, get filled
- All states above Fermi Energy Level (μ) remains empty
- In book Fermi Energy Level is represented as E_F instead of μ

- At 0K, Fermi Distribution Function $f(E)$ is of the form of a unit step function: $u(\mu - E)$

Fermi Distribution Function: Non-Zero Temperature

- At higher temperature, some electrons possess thermal energy more than Fermi Energy μ
- The distribution around μ gets blurred
- The exact distribution can be derived using Statistical Thermodynamics

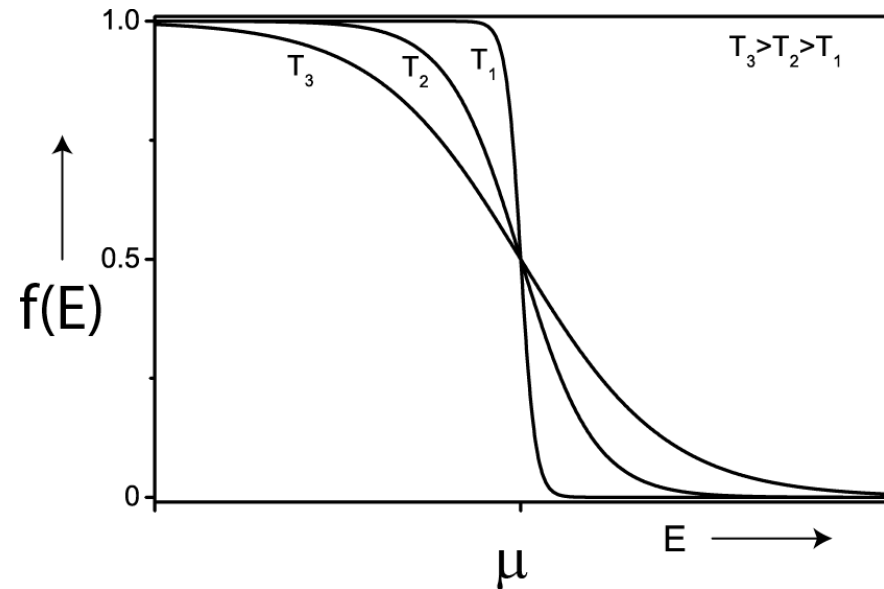
Fermi Distribution Function:

$$f(E) = \frac{1}{e^{\frac{E-\mu}{kT}} + 1}$$

$f(E)$: Probability that a state with energy E will be occupied

μ : Fermi Energy Level (Electrochemical Potential)

k : Boltzmann constant = $1.381 \times 10^{-23} \frac{J}{K} \approx 80 \mu eV/K$



At higher temperature, the transition from filled states to empty states is more gradual

- Transition is largely over an energy range of $\sim \pm 2kT$ around $E = \mu$, where k is the Boltzmann constant and T is the absolute temperature

Fermi Distribution Function: Properties

- Fermi Distribution Function:

$$f(E) = \frac{1}{e^{\frac{E-\mu}{kT}} + 1}$$

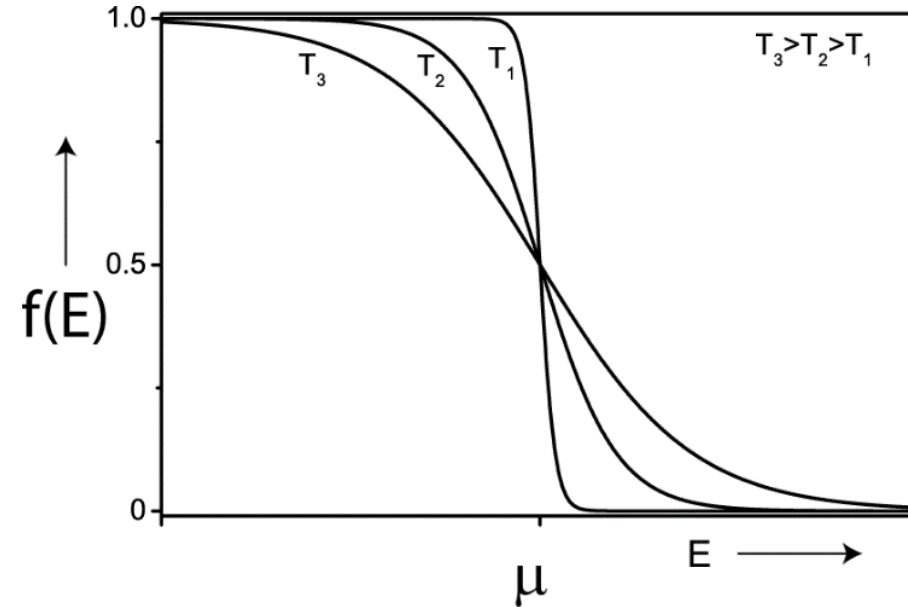
1. $0 \leq f(E) \leq 1$

2. $f(E = \mu) = \frac{1}{2}$

3. For $(E - \mu) \gg kT$

$f(E) = e^{-\frac{E-\mu}{kT}} \ll 1$ [Maxwell-Boltzmann Distribution]

4. Low energies are always occupied, high energies are always empty



Intrinsic Semiconductor in Equilibrium

Fermi Distribution Function: Application

Probability that an energy level E is occupied by an electron at equilibrium $\equiv$ **Fermi Distribution Function** $\equiv f(E)$

Number of states in an interval E and $E + dE$: $D(E)dE$

Number of electrons in an interval E and $E + dE$: $dN = f(E)D(E)dE$

This can be used to compute the number of electrons in a semiconductor within some energy range: E_1 and E_2

Equilibrium Electron Concentration

Problem:

Find the concentration of electrons (number per unit volume) in the conduction band in 3-D bulk material for which density of states is given as:

$$g_c(E) = \frac{8\pi m_n^* \sqrt{2m_n^* (E - E_c)}}{h^3}$$

Assume that:

The start of the conduction band is at energy $E = E_c$ and $E_c - \mu \gg kT$ [Maxwell-Boltzmann Statistics can be used]

Use Gamma Function for integration:

$$\int_0^\infty \sqrt{x} e^{-x} dx = \Gamma\left(\frac{3}{2}\right) = \frac{\sqrt{\pi}}{2}$$

Answer:

$$N = \frac{4\sqrt{2}(\pi m_n^* kT)^{3/2}}{h^3} e^{-(E_c - \mu)/kT}$$

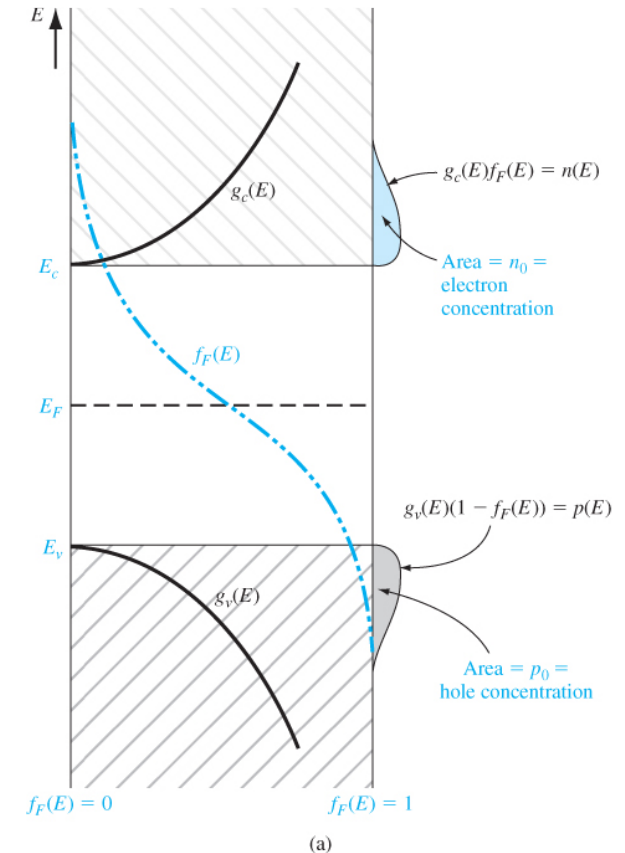


Figure 4.1 | (a) Density of states functions, Fermi-Dirac probability concentrations for the case when E_F is near the midgap energy; (b) and (c) expanded view near the valence-band energy.

Effective DOS (Conduction Band)

- Thermal equilibrium concentration of electrons:

$$n_0 = \frac{4\sqrt{2}(\pi m^* kT)^{3/2}}{h^3} e^{-(E_C - E_F)/kT}$$

- The factor $N_C = \frac{4\sqrt{2}(\pi m^* kT)^{3/2}}{h^3}$ is known as Effective DOS in the conduction band.

- The effective DOS $N_C = \frac{4\sqrt{2}(\pi m^* kT)^{3/2}}{h^3}$ depends on effective mass of electron and temperature

- If effective mass of electron is taken as the rest mass and $T = 300\text{ K}$,

$$\text{effective DOS } N_C = 2.5 \times 10^{19} \text{ cm}^{-3}$$

Equilibrium Hole Concentration

Problem:

Find the concentration of holes (number per unit volume) in the conduction band in 3-D bulk material for which density of states is given as:

$$g_V(E) = \frac{8\pi m_p^* \sqrt{2m_p^* (E_V - E)}}{h^3}$$

Assume that:

The start of the conduction band is at energy $E = E_V$ and $(\mu - E_V) \gg kT$ [Maxwell-Boltzmann Statistics can be used]

Use Gamma Function for integration:

$$\int_0^\infty \sqrt{x} e^{-x} dx = \Gamma\left(\frac{3}{2}\right) = \frac{\sqrt{\pi}}{2}$$

Answer:

$$N = \frac{4\sqrt{2}(\pi m_p^* kT)^{3/2}}{h^3} e^{-(\mu - E_V)/kT}$$

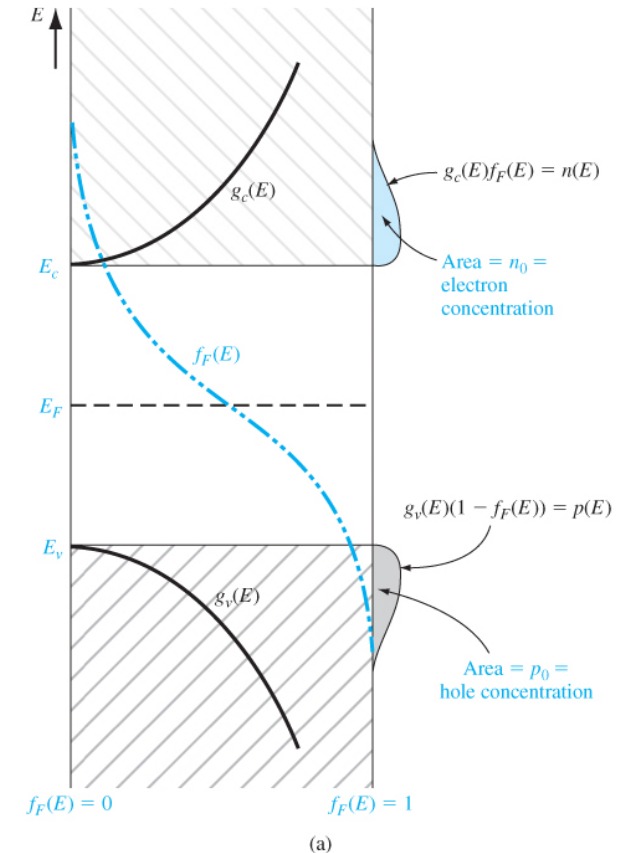


Figure 4.1 | (a) Density of states functions, Fermi-Dirac probability concentrations for the case when E_F is near the midgap energy; (b) and (c) expanded view near the valence-band energy.

Effective DOS (Valence Band)

- Thermal equilibrium concentration of holes:

$$p_0 = \frac{4\sqrt{2}(\pi m_p^* kT)^{3/2}}{h^3} e^{-(E_F - E_V)/kT}$$

- The factor $N_V = \frac{4\sqrt{2}(\pi m_p^* kT)^{3/2}}{h^3}$ is known as Effective DOS in the valence band.

- The effective DOS $N_V = \frac{4\sqrt{2}(\pi m_p^* kT)^{3/2}}{h^3}$ depends on effective mass of holes and temperature
- At $T = 300\text{ K}$, effective DOS N_V is of the order of 10^{19} cm^{-3}

Table 4.1 | Effective density of states function and density of states effective mass values

	$N_c\text{ (cm}^{-3}\text{)}$	$N_v\text{ (cm}^{-3}\text{)}$	m_n^*/m_0	m_p^*/m_0
Silicon	2.8×10^{19}	1.04×10^{19}	1.08	0.56
Gallium arsenide	4.7×10^{17}	7.0×10^{18}	0.067	0.48
Germanium	1.04×10^{19}	6.0×10^{18}	0.55	0.37

Intrinsic Carrier Concentration (1)

- Thermal equilibrium concentration of electrons:

$$n_0 = N_C e^{-(E_C - E_F)/kT}$$

- Thermal equilibrium concentration of holes:

$$p_0 = N_V e^{-(E_F - E_V)/kT}$$

- At thermal equilibrium concentration of electrons and holes are same (material is neutral)
- We call this concentration as intrinsic carrier concentration and denote as n_i

$$\Rightarrow n_i = n_0 = p_0$$

Table 4.1 | Effective density of states function and density of states effective mass values

	$N_c \text{ (cm}^{-3}\text{)}$	$N_v \text{ (cm}^{-3}\text{)}$	m_n^*/m_0	m_p^*/m_0
Silicon	2.8×10^{19}	1.04×10^{19}	1.08	0.56
Gallium arsenide	4.7×10^{17}	7.0×10^{18}	0.067	0.48
Germanium	1.04×10^{19}	6.0×10^{18}	0.55	0.37

- We can write:

$$\begin{aligned}
 n_0 \times p_0 &= n_i^2 \\
 &= N_C e^{-(E_C - E_F)/kT} \times N_V e^{-(E_F - E_V)/kT} \\
 &\Rightarrow n_i^2 = N_C N_V e^{-(E_C - E_V)/kT} \\
 &\Rightarrow n_i^2 = N_C N_V e^{-E_G/kT}
 \end{aligned}$$

Intrinsic Carrier Concentration (2)

$$n_i^2 = N_C N_V e^{-E_G/kT}$$

Table 4.2 | Commonly accepted values of n_i at $T = 300$ K

Silicon	$n_i = 1.5 \times 10^{10} \text{ cm}^{-3}$
Gallium arsenide	$n_i = 1.8 \times 10^6 \text{ cm}^{-3}$
Germanium	$n_i = 2.4 \times 10^{13} \text{ cm}^{-3}$

Extrinsic Semiconductor in Equilibrium

Intrinsic vs. Extrinsic Semiconductor

- Intrinsic semiconductor:

- pure single-crystal semiconductor material
- At equilibrium, electron and hole concentrations balanced

- Extrinsic semiconductor:

- semiconductor material with donor or acceptor atoms
- an imbalance of electrons/holes

Doping with Donors (1)

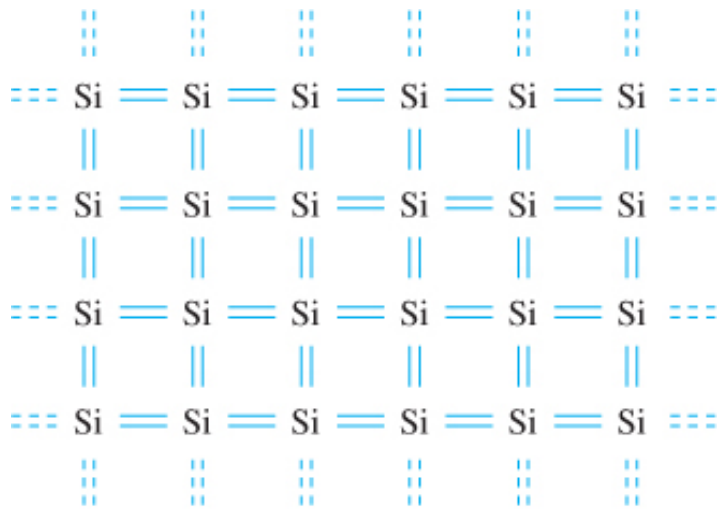


Figure 4.3 | Two-dimensional representation of the intrinsic silicon lattice.

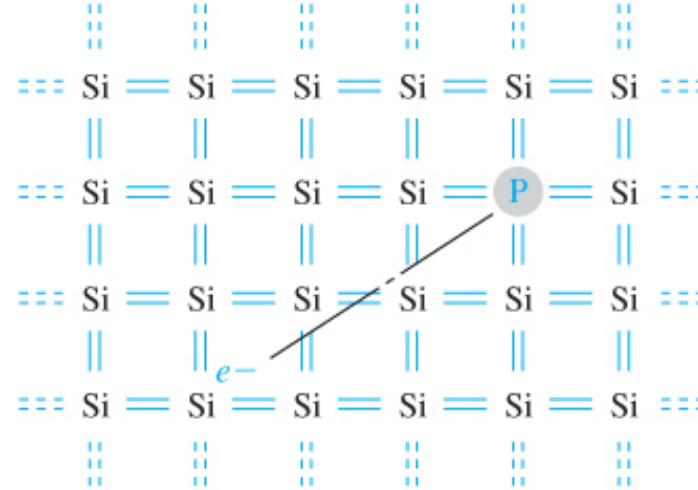


Figure 4.4 | Two-dimensional representation of the silicon lattice doped with a phosphorus atom.

- Doping with phosphorous atom (has five valence electrons)
- Four electrons form covalent bonds and the fifth electron is loosely bound to Phosphorous atom

- At $T=0$ K fifth electron bound to Phosphorous atom
- But it can jump to conduction band by small thermal energy (elevating the temperature)

Doping with Donors (2)

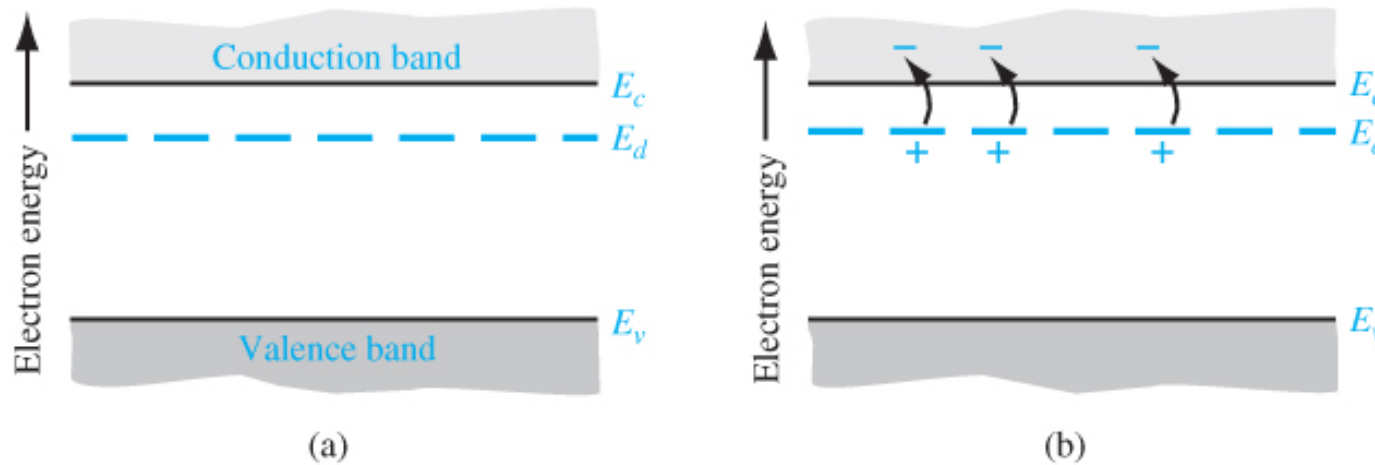


Figure 4.5 | The energy-band diagram showing (a) the discrete donor energy state and (b) the effect of a donor state being ionized.

- The energy state of donor electron is shown as E_d
- It is close to the conduction band

- By gaining small amount of energy it can jump to the conduction band

- Free electrons becomes available in the conduction band for current flow
- Phosphorous atom becomes positively charged phosphorous ion (which does not move and remains fixed in the lattice)

- No holes are created, thus there is an excess of electrons in the material
- This type of material is known as ***n-type semiconductor***

Doping with Acceptors (1)

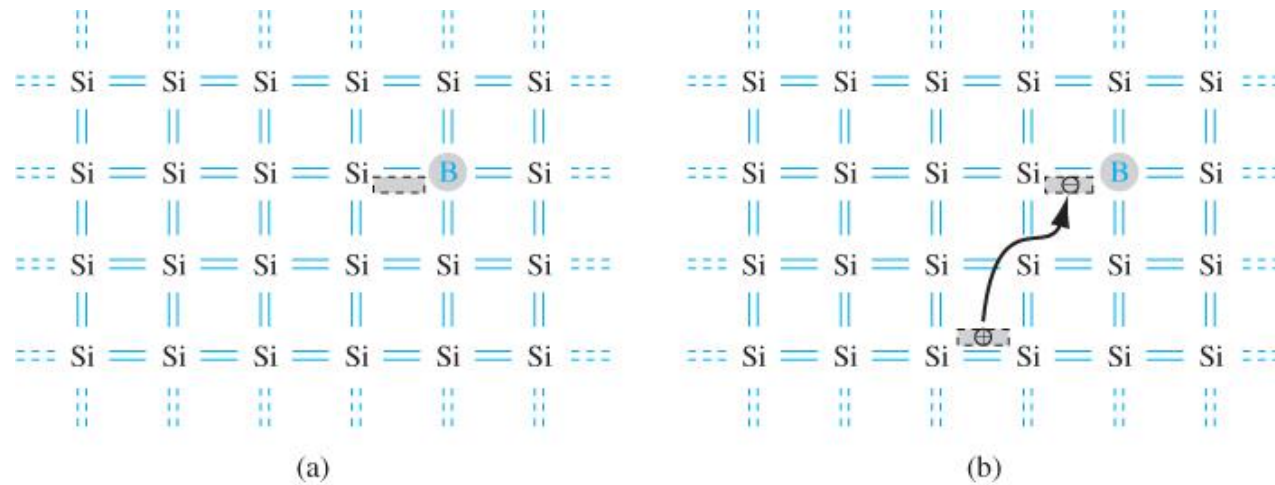


Figure 4.6 | Two-dimensional representation of a silicon lattice (a) doped with a boron atom and (b) showing the ionization of the boron atom resulting in a hole.

- Doping with Boron atom (has three valence electrons)
- Three electrons form three covalent bonds and the fourth covalent bond has an empty state for the electron

- At $T=0$ K fourth covalent is empty (and Boron atom is neutral)
- At elevated temperature, electron from other covalent bonds (in valence band) can jump due to small thermal energy to make the fourth covalent bond complete
 - Leaves a hole in the valence band

Doping with Acceptors (2)

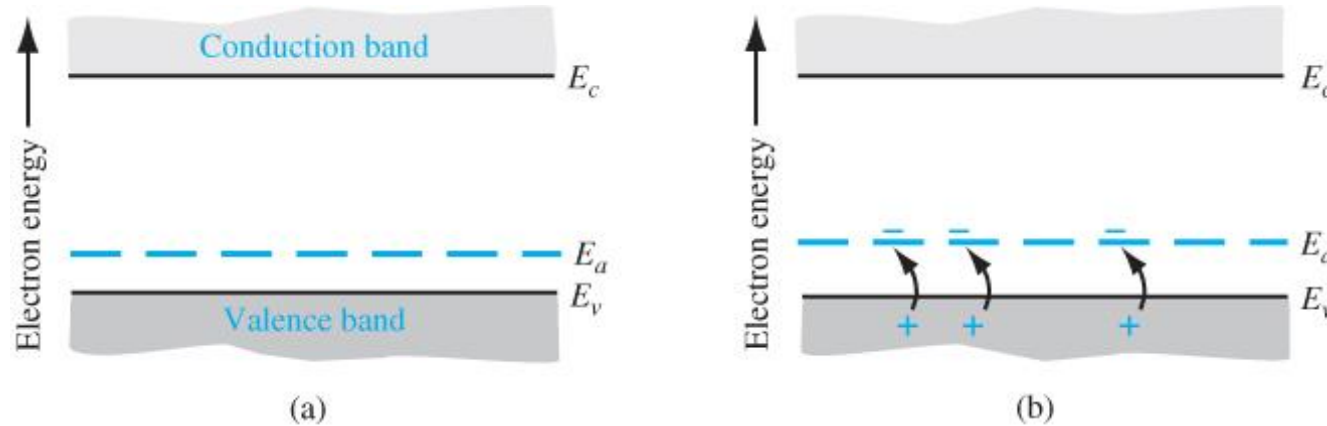


Figure 4.7 | The energy-band diagram showing (a) the discrete acceptor energy state and (b) the effect of an acceptor state being ionized.

- The energy state of acceptor state is shown as E_a
- It is close to the valence band
- By gaining small amount of energy electrons can jump to the acceptor state from the valence band

- Holes becomes available in the valence band for current flow
- Boron atom becomes negatively charged boron ion (which does not move and remains fixed in the lattice)

- No free electrons are created, thus there is an excess of holes in the material
- This type of material is known as ***p-type semiconductor***

Dopants and Ionization

- Ionization Energy: Energy required for the:
 - Electron to jump from the donor state to conduction band (and create a free electron)
 - Electron to jump from the valence band to the acceptor state (and create a free hole)

Table 4.3 | Impurity ionization energies in silicon and germanium

Impurity	Ionization energy (eV)	
	Si	Ge
<i>Donors</i>		
Phosphorus	0.045	0.012
Arsenic	0.05	0.0127
<i>Acceptors</i>		
Boron	0.045	0.0104
Aluminum	0.06	0.0102

For GaAs:

- Donor: Group VI (substituting As), Group IV (replacing Ga)
- Acceptor: Group II, (substituting Ga) Group IV (substituting As)

Table 4.4 | Impurity ionization energies in gallium arsenide

Impurity	Ionization energy (eV)
<i>Donors</i>	
Selenium	0.0059
Tellurium	0.0058
Silicon	0.0058
Germanium	0.0061
<i>Acceptors</i>	
Beryllium	0.028
Zinc	0.0307
Cadmium	0.0347
Silicon	0.0345
Germanium	0.0404

References

- D. A. Neaman, "*Semiconductor Physics and Devices*"